

KAHBOD AEINI

Calgary, AB (Relocating to Waterloo, Jan 2026)
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SUMMARY

Graduate researcher in Computer Science with expertise in machine learning, distributed systems, and data-driven algorithms. Skilled in Python, C++, and modern ML frameworks (PyTorch, TensorFlow, scikit-learn). Experienced in applied ML projects spanning bioinformatics, NLP, and data engineering. Eager to contribute to cutting-edge AI research and its real-world applications.

EDUCATION

University of Waterloo PhD in Electrical and Computer Engineering (incoming)	Jan 2026 –
University of Calgary Master of Computer Science (GPA: 3.8/4.0)	2023 – 2025
Sharif University of Technology Bachelor of Computer Engineering	2019 – 2023

EXPERIENCE

Research Assistant, Johns Hopkins University (Remote)	Feb 2023 – Aug 2023
- Built ML models in Python/R to predict immune cell composition from lung cancer transcriptomics. - Applied feature selection, classification, and enrichment analysis to biological datasets.	
Data Engineer, AI Lida	Feb 2022 – Sep 2022
- Built and optimized ETL pipelines using Python and PostgreSQL, improving data throughput and reliability. - Collaborated with the Security team to design secure data workflows for production systems. - Hired, trained, and mentored 5 data engineering interns.	
Research Assistant, University of Calgary	Jun 2024 – Present
- Designed and analyzed algorithms for concurrency and synchronization in distributed systems. - Proved space- and time-complexity bounds for randomized wait-free locks in shared-memory models. - Applied advanced probability and complexity theory to evaluate performance guarantees.	

PUBLICATIONS

Kahbod Aeini, Dante Bencivenga, George Giakkoupis, Philipp Woelfel.
Simple and Efficient Randomized Wait-Free Lock.
Submitted to ACM-SIAM Symposium on Discrete Algorithms (SODA 2026), under review.

PROJECTS

- Fraud Detection System (BSc Thesis)**

Developed and compared 3 machine learning models for detecting fraudulent banking transactions.

GitHub
- Information Retrieval System (NLP + SWD)**

Designed and implemented a retrieval system for twitter interaction and health articles with GUI, Classification and Clustering tweets and articles.

GitHub
- Users Profile Matching (Computer Vision)**

Complete implementation and summarization of “User profiles matching for different social networks based on faces identification” paper as the final project of the Numerical Computation course.

GitHub
- Transcriptomics Study (Bioinformatics)**

Transcriptomics study of AML to identify differentially expressed genes and associated pathways, potentially causal for the disease.

GitHub
- Secure Chat Application**

Built a Python-based secure messaging app with encryption features for safe communication.

GitHub

AWARDS

Departmental Research Award, University of Calgary (2023)
Top 100 National Universities Entrance Exam, Iran (2019)
National Elites Fellowship (2019)

SKILLS

ML/AI: PyTorch, TensorFlow, scikit-learn, NLP, GCNs
Programming: Python, C/C++, Java, R, Haskell, Prolog
Data: PostgreSQL, MongoDB, MySQL, Pandas, NumPy
Tools: Git, Linux, Docker, NixOS, Wordpress

RELEVANT COURSES

Randomized Algorithms, Data Mining & Machine Learning, Modern Information Retrieval, Artificial Intelligence, Database Design, Bioinformatics, Numerical Computation, Linear Algebra